

CONTEMPORARY REVIEW

Catecholaminergic Polymorphic Ventricular Tachycardia: Advancing From Molecular Insights to Preclinical Models

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ABSTRACT: Inherited cardiac channelopathies are linked to a heightened risk of sudden cardiac death. Despite evolving knowledge on different genes for these inherited conditions, for certain subtypes, such as catecholaminergic polymorphic ventricular tachycardia syndrome, the specific genetic causes remain unidentified. The research of the pathophysiological mechanisms underlying catecholaminergic polymorphic ventricular tachycardia syndrome has been conducted through different in vitro and in vivo models, including genetically modified animal models, cardiac-specific transgenic models, pharmacological interventions in animal models, human-induced pluripotent stem cell-derived cardiomyocytes in 2- and 3-dimensional cardiac models. Recent research predominantly utilizes human-induced pluripotent stem cell-derived cardiomyocytes, focusing on genotype–phenotype correlations and pharmacological screening. The integration of cutting-edge techniques such as clustered regularly interspaced short palindromic repeats/Cas9 genome editing and 3-dimensional-engineered heart tissues has shed new light on the pathophysiological mechanisms of catecholaminergic polymorphic ventricular tachycardia, potentially enhancing drug therapies as part of personalized medicine approaches. This review emphasizes the diverse insights gained from both in vivo and in vitro studies of catecholaminergic polymorphic ventricular tachycardia, along with the application of these models in various research contexts.

Key Words: 3-dimensional engineered heart ■ catecholaminergic polymorphic ventricular tachycardia ■ CRISPR/Cas9 genome editing ■ electrophysiological phenotyping ■ human-induced pluripotent stem cell-derived cardiomyocytes ■ precision medicine in cardiac channelopathies ■ preclinical cardiovascular models

CATECHOLAMINERGIC POLYMORPHIC VENTRICULAR TACHYCARDIA IN CLINICAL PERSPECTIVE

Epidemiology and Genetic Recognition of Catecholaminergic Polymorphic Ventricular Tachycardia

Catecholaminergic polymorphic ventricular tachycardia (CPVT) was first recognized as a unique inherited arrhythmogenic syndrome in 1975.¹ In 1999, an autosomal dominant pattern of inheritance was

described, with its first genetic locus located on chromosome 1q42-43.² Most CPVT cases are associated with variants in the *RYR2* (ryanodine receptor 2) gene.³ Triggered by emotional stress or physical exertion, CPVT manifests in individuals without structural heart disease, leading to adrenergic-induced bidirectional or polymorphic ventricular tachycardia (Figure 1A).⁴ Its prevalence in Europe is estimated at 1 in 10000 and can result in sudden cardiac death. Children and adolescents are commonly affected, with a notably poor prognosis for those experiencing early syncope.⁵ Untreated, CPVT has a high mortality rate, with up to 31% of patients dying by the age of 30.⁶

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Nonstandard Abbreviations and Acronyms

CASQ2	cardiac calsequestrin 2
CPVT	catecholaminergic polymorphic ventricular tachycardia
CRISPR	clustered regularly interspaced short palindromic repeats
DAD	delayed afterdepolarization
EHT	engineered heart tissue
hiPSC-CM	human-induced pluripotent stem cell-derived cardiomyocyte
iPSC-CM	induced pluripotent stem cell-derived cardiomyocyte
RyR2	ryanodine receptor 2
SAN	sinoatrial node
SR	sarcoplasmic reticulum
TECRL	trans-2,3-enoyl-CoA reductase-like protein

CURRENT TREATMENT APPROACHES

Treatment of CPVT involves a comprehensive approach that emphasizes symptoms without clear evidence from large-scale trials. β Blockers, particularly those without sympathomimetic effects, are first-line treatments to reduce syncope and control arrhythmias.⁷ However, managing CPVT can be complex, especially when patients exhibit ventricular tachycardia during exercise.⁸ For those inadequately responsive to β blockers, calcium channel blockers like diltiazem or verapamil have been found to stabilize heart rhythms.⁹ Additionally, flecainide, combined with β blockers, can decrease the risk of ventricular tachyarrhythmias.¹⁰ It is essential to highlight that the efficacy of these drugs can vary among patients with CPVT with different gene variants, and it is not yet fully understood if these treatments are universally effective across all CPVT types. Patients with acute cases of CPVT and left cardiac sympathetic denervation surgery may be needed to regulate sympathetic nerve activity and reduce arrhythmias.⁵ Concurrently, patients at increased risk for sudden cardiac death may require the implantation of an implantable cardioverter-defibrillator to prevent fatal arrhythmic events.¹¹ Lifestyle adaptations are crucial in CPVT management. Avoidance of strenuous physical activities and intense emotional stress, known triggers of arrhythmic episodes, is advised.⁵ The European Heart Rhythm Association discourages participation in competitive and high-intensity sports for patients with CPVT, recommending low- to moderate-intensity

activities under certain conditions, like passing a stress test and remaining symptom free for a minimum of 3 months. The European Heart Rhythm Association advises asymptomatic CPVT variant carriers to engage in low-intensity sports and suggests that patients with CPVT adopt an exercise routine similar to the general population's standards.¹² In line with the 2022 European Society of Cardiology guidelines, there is now an increased emphasis on genetic testing and cardiac magnetic resonance imaging for risk stratification in patients with CPVT, and the guidelines advocate for personalized management plans.¹³

CPVT'S MOLECULAR PUZZLE AND PATHOPHYSIOLOGY

The variants in genes for RYR2 and CASQ2 (cardiac calsequestrin 2) define CPVT.¹⁴ RYR2, a tetrameric protein on the sarcoplasmic reticulum (SR) membrane, facilitates ion release during systole (Figure 1).¹⁵ It is tethered to CASQ2 through satellite proteins.¹⁶ Associated with RYR2's cytoplasmic region are calmodulin, FKBP12.6 (calstabin2), protein kinase A, phosphatase 1, and phosphatase 2, while triadin links to the luminal side.¹⁷ Crucial to CPVT pathophysiology is calcium homeostasis. Variants in RYR2 and CASQ2 disrupt cardiac cellular calcium signaling. These alterations in calcium regulation lead to atypical Ca^{2+} release under sympathetic stimulation.¹⁸

The genetic heterogeneity of CPVT can be further understood by distinguishing between its 2 main subtypes: CPVT1 and CPVT2. CPVT1 is primarily associated with mutations in the gene encoding the RYR2. Mutations in this gene lead to abnormal calcium release from the SR. CPVT2, on the other hand, is linked to mutations in the gene encoding CASQ2. This gene is crucial for calcium storage within the SR. Mutations in CASQ2 disrupt calcium homeostasis, similarly, leading to arrhythmogenic events.¹⁹ Both CPVT1 and CPVT2 variants typically cause SR calcium leakage, triggering delayed afterdepolarizations (DADs) via cytosolic calcium accumulation (Figure 1B). The released Ca^{2+} activates the $\text{Na}^+/\text{Ca}^{2+}$ exchanger, producing transient inward currents and potential DADs, heightening arrhythmia risk. Conversely, early afterdepolarizations emerge during action potential plateaus or repolarization, overlapping with CPVT symptoms. Apart from RYR2 gene variants, CPVT also involves less common genetic changes in triadin, calmodulin genes (*CALM1*, *CALM2*, *CALM3*), ankyrin-B (*ANK2*), and Trans-2,3-enoyl-CoA reductase-like protein (*TECRL*), a gene involved in calcium homeostasis and linked to an autosomal recessive form of CPVT (Figure 1B).²⁰ However, the role of *KCNJ2*, *PKP2*, and *SCN5A* genes in CPVT is now primarily discounted, with the prevalent *ANK2* gene

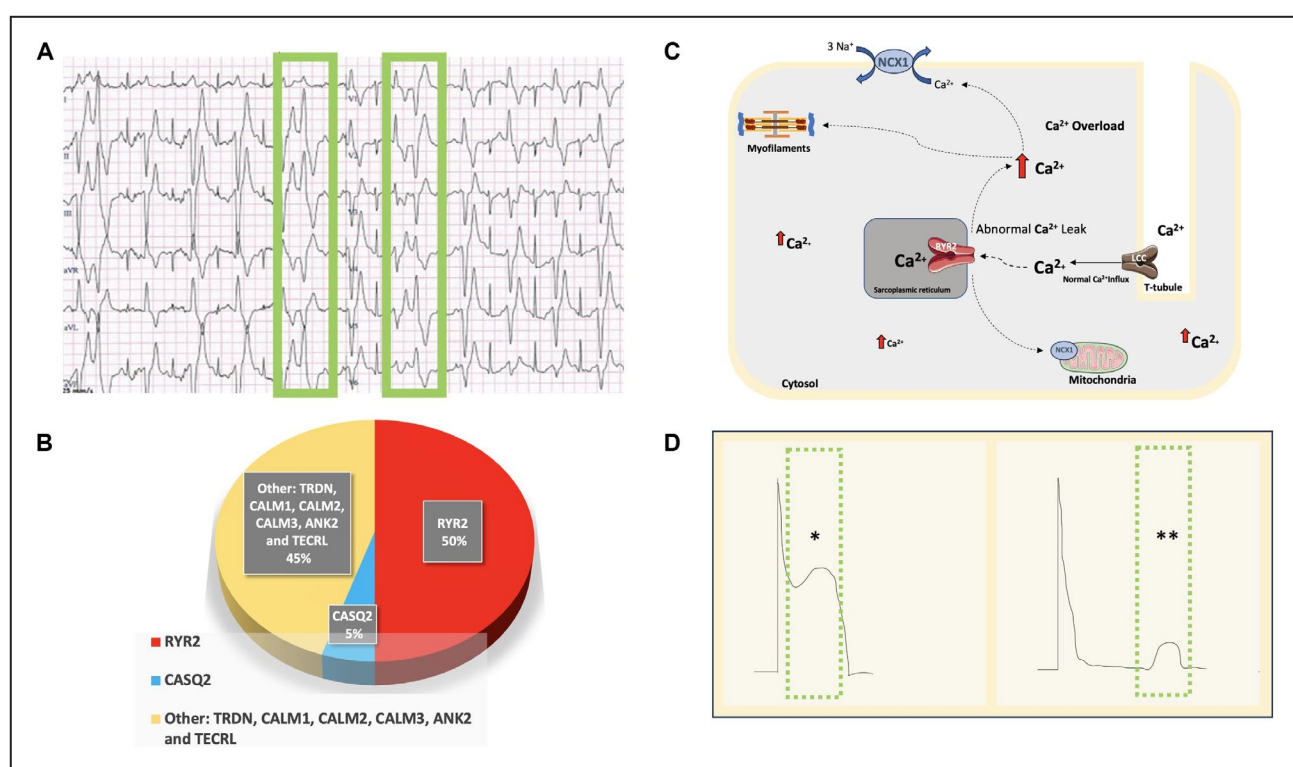


Figure 1. Main characteristics and critical mechanism of CPVT.

A, Typical ECG of a patient with CPVT, characterized by abrupt sequences of polymorphic ventricular complexes that punctuate the baseline rhythm. **B**, Comprehensive listing of all reported genes related to CPVT. **C**, Calcium dynamics in a CPVT cardiomyocyte: Normal Ca^{2+} influx via L-type Ca^{2+} channels during depolarization (systole) triggers RYR2 to release Ca^{2+} from the sarcoplasmic reticulum. In contrast, RYR2 mutations induce an abnormal Ca^{2+} leak during repolarization (diastole), resulting in Ca^{2+} overload. The NCX1 attempts to expel excess Ca^{2+} from the cytosol by exchanging it for Na^{+} . This generates an inward Na^{+} current, contributing to delayed afterdepolarizations. **D**, *Early after depolarization and **delayed after polarization. Figure 1C created using Smart Servier Medical Art. CPVT indicates catecholaminergic polymorphic ventricular tachycardia; NCX1, $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger; and RYR2, cardiac ryanodine receptor.

variant unlikely to be a primary CPVT factor.²⁰ In 2016, the *TECRL* mutation's link to CPVT was first reported.²¹ Recent studies on *TECRL* knockout mice also indicate mitochondrial function alterations.²² Pathomechanisms for CPVT include theories like weakened RYR2 and FKBP12.6 binding, causing FKBP12.6 dissociation and channel leakage during diastole.²³ The store-overload-induced Ca^{2+} release hypothesis suggests a lowered store-overload-induced calcium release threshold in RYR2 mutants, causing SR Ca^{2+} overflow.²⁴ Another perspective involves defective intramolecular domain interaction, leading to SR Ca^{2+} leakage.²⁵ Additionally, patients with CPVT may experience ventricular fibrillation. This can arise from irregular automaticity of Purkinje fibers. Such irregularities lead to abnormalities in cardiac impulse propagation.²⁶ Elevated catecholamine levels, a key trigger for ventricular arrhythmias in CPVT, amplify the response to sympathetic stimulation.²⁷ Variants affecting the β -adrenergic signaling pathway and overactivity of the calmodulin-dependent kinase II pathway in individuals with ryanodine receptor variants are also implicated in CPVT.²⁸

EXPERIMENTAL MODELS OF CPVT

The study of CPVT has been dramatically advanced using both in vivo and in vitro models, each offering unique insights into the disease's mechanisms and potential treatments (Figure 2). In vivo models are crucial for understanding CPVT's systemic physiological and pathophysiological responses. These models allow researchers to observe the disease's progression and response to therapies in a complex living system, closely mirroring the human condition (Table 1).^{29–36}

ANIMAL MODELS IN CPVT RESEARCH

Mohamed et al discuss using transgenic knock-in mouse models to elucidate the pathophysiological mechanisms underlying CPVT. These models harbor specific mutations in the RYR2 or CASQ2 genes, which correlate with the autosomal dominant and recessive forms of CPVT, respectively. The research

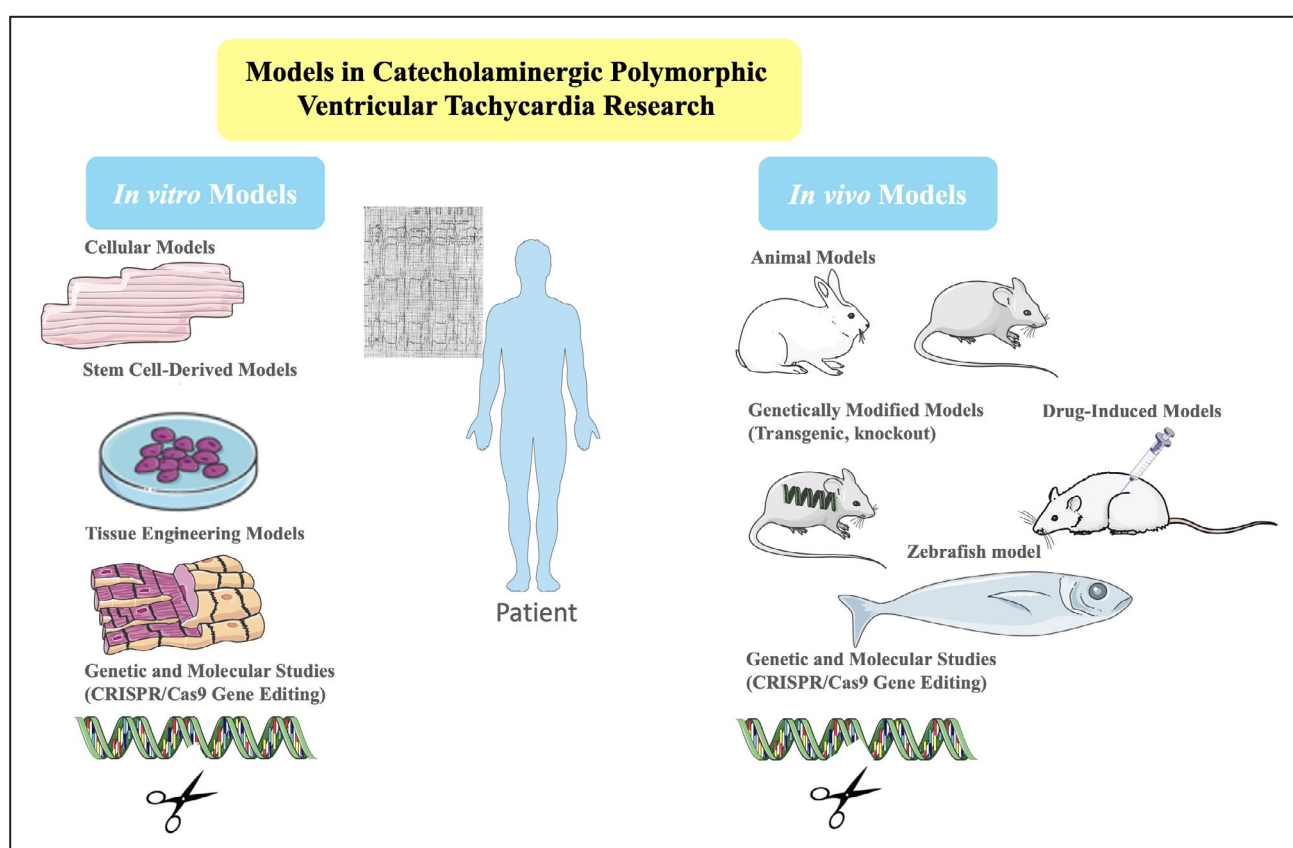


Figure 2. Overview of in vivo and in vitro models in CPVT research.

This figure illustrates the main models used to investigate CPVT pathophysiological mechanisms. In vivo models include animal models such as rodents and rabbits, genetically modified animals that mimic the human disease phenotype, enabling the study of disease progression and therapeutic interventions, and drug-induced models to probe the electrophysiological properties and response to therapeutic interventions. In vitro models encompass cellular constructs, iPSC-CMs, and tissue engineering approaches. Both in vivo and in vitro models benefit from genetic and molecular studies such as CRISPR/Cas9 gene editing, which help elucidate gene function and validate therapeutic targets. The schematic also represents the translation from clinical ECG diagnosis to research model application, highlighting the bidirectional flow of insights between patient-derived data and experimental findings. Image created using Smart Servier Medical Art. CPVT indicates catecholaminergic polymorphic ventricular tachycardia; CRISPR, clustered regularly interspaced short palindromic repeats; and iPSC-CMs, induced pluripotent stem cell-derived cardiomyocytes.

demonstrated that CPVT-associated arrhythmias originate from DADs and triggered activities, providing insight into the molecular and electrophysiological basis of the disease.^{37,38} Recent advances also highlight the use of zebrafish as an in vivo model for inheritable ventricular arrhythmia syndromes. Zebrafish share remarkable similarities with humans in cardiac electrophysiology, and advanced gene editing and electrophysiological phenotyping methods are available.³³

Molecular Mechanisms and Calcium Handling in Animal Models

Diving deeper into molecular mechanisms, Veress et al provided insights into the role of endoplasmic reticulum stress in contributing to cardiac arrhythmias CPVT.³⁸ Using a rat model of CPVT induced by the RyR2-S222L (+/-) variant, they observed that CPVT ventricular myocytes under β -adrenergic stimulation

exhibit a heightened propensity to spontaneous Ca^{2+} waves and DADs. Further exploring molecular interactions, Veress et al also explored the diverse arrhythmia phenotypes caused by mutations in *CALM2* and *CALM3*. Their research revealed that, while mutations in *CALM2* lead to a mixed phenotype resembling that in CPVT, mutations in *CALM3* result in typical CPVT characteristics. They demonstrated that both *CALM2* D132E and *CALM3* D132E rats exhibit defective intracellular Ca^{2+} handling, impacting the function of both RYR2 and L-type calcium channels. Shifting the focus to cellular metabolism, a recent study has shed light on the role of mitochondria in mitigating the harmful effects of aberrant Ca^{2+} release in CPVT. Mitochondrial phosphate transport, known to influence mitochondrial calcium uptake and buffering, is hypothesized to impact arrhythmogenesis in CPVT. In experiments using a CPVT mouse model (CASQ2 knockout), modulation of mitochondrial phosphate transport through

Table 1. Experimental Models for CPVT

	Model	Methods	Advantages	Disadvantages
Animal models	Transgenic mouse model (<i>RYR2</i> variant) ²⁹	Creating transgenic models where specific genes are introduced or overexpressed	Direct observation of pathophysiological effects; suitable for long-term studies	Limited by differences in cardiac physiology compared with humans
	Genetically modified animal model ³⁰	It encompasses various genetic modifications including knock-ins, knockouts, and transgenics	Allows exploration of gene function and interaction in vivo; more versatile than specific transgenic models	Ethical considerations; may not fully replicate human disease mechanisms
	Rat models ³¹	Transgenic expression, pharmacological induction	Allows study of specific protein interactions; useful for drug testing	May not fully replicate human arrhythmic triggers
	Rabbit model (sex differences) ³²	Electrophysiological studies on ion currents and calcium handling	Insight into sex-specific electrophysiological differences; applicable to understanding human sex differences	Complex handling and transgenic model are difficult to obtain
	Zebrafish model ³³	CRISPR/Cas9 for targeted gene mutations	Rapid development, ease of genetic manipulation; translucency allows direct observation of cardiac function	Cardiac anatomy and function may not fully replicate human conditions
hiPSC-CMs	hiPSCs-CM models ³⁴	CRISPR/Cas9 for gene editing, calcium imaging, patch-clamping	Patient-specific modeling; high relevance to human cellular responses	Lack of cardiac markers; complexity in differentiation and maturation; high variability
	3-dimensional cardiac tissue model ³⁵	Tissue engineering with iPSC-CMs, mechanical and electrical stimulation	Mimics in vivo microenvironment; integrates cellular interactions and mechanics	Technically challenging; expensive to develop and maintain
Computational models	Computational simulation ³⁶	Simulation of ion channel dynamics and cardiac electrophysiology	Hypothesis testing without ethical constraints; integrates vast data sets to predict outcomes	Depends on model accuracy; may not mirror all biological complexities

CPVT indicates catecholaminergic polymorphic ventricular tachycardia; CRISPR, clustered regularly interspaced short palindromic repeats; hiPSC-CM, human-induced pluripotent stem cell–derived cardiomyocyte; and iPSC-CM, induced pluripotent stem cell–derived cardiomyocyte.

pharmacological inhibition or manipulation of cytosolic phosphate concentration significantly affected cellular calcium dynamics and arrhythmic burden.³⁹ Examining sex-specific aspects, Horváth et al investigated sex differences in Na^+ and Ca^{2+} handling properties in rabbit atrial cardiomyocytes.³² In their study using adult male and female rabbits and female rabbits treated with dihydrotestosterone, they assessed ion currents and calcium-handling properties to investigate potential sex differences in atrial electrophysiology. Additionally, the study observed that females generally exhibited larger forward $\text{Na}^+/\text{Ca}^{2+}$ exchanger currents than males. Moreover, calcium transient measurements showed that females had higher diastolic Ca^{2+} and Ca^{2+} transient amplitude. This study provides valuable insights into sex-specific differences in cardiac electrophysiology in CPVT, which could inform our understanding of conditions like CPVT and guide sex-specific therapeutic strategies. Further contributing to our molecular understanding, the study by Lin et al studied the effects of *TECL* deficiency on cardiometabolic and Ca^{2+} regulation in cardiac tissue, shedding light on potential mechanisms underlying CPVT.⁴⁰ They establish *TECL* knockout mice and conduct RNA-sequencing analysis, identifying 857 differentially expressed genes

in *TECL* knockout mice. Advancing the discussion to β -adrenergic receptor signaling, Pavlaki et al used mouse models to selectively disrupt integral components of β -adrenergic receptor signaling and gain mechanistic insight into CPVT pathogenesis.⁴¹ They found that abnormal Ca^{2+} release events induced by hyperactivated L-type calcium channels contributed to arrhythmia incidence, suggesting a link between β -adrenergic receptor stimulation, L-type calcium channel-mediated calcium increase, and CPVT development. The observed change in L-type calcium channel activity constitutes pathogenic hyperactivation, not merely an increase in current flow. Investigating sinoatrial node (SAN) dysfunction, Neco et al investigated the SAN dysfunction in mice carrying the CPVT-linked *RyR2R4496C* variant.⁴² Their findings revealed that these mice exhibited impaired SAN automaticity following β -adrenergic stimulation, mirroring the SAN dysfunction seen in CPVT patients after exercise. The study observed slower pacemaker activity, reduced chronotropic response in *RyR2R4496C* SAN cells, and increased frequency of Ca^{2+} sparks. Interestingly, they found that this variant led to a paradoxical decrease in SAN automaticity due to a Ca^{2+} -dependent reduction of the L-type Ca^{2+} current and sarcoplasmic

reticulum Ca^{2+} depletion during diastole. Following up on the SAN dysfunction observed in the study by Neco et al., the long-term findings from Veith et al provide crucial clinical context. Despite β blocker use and additional flecainide when needed, patients with CPVT still faced a high rate of ventricular tachycardia and fibrillation.⁴³ In an interesting contribution, the study by Yin et al investigated the arrhythmogenic mechanisms of a mutation in the *RYR2* N-terminal domain associated with CPVT.⁴⁴ They observed ventricular tachycardias in both patients with CPVT and knock-in mice harboring the variant, along with increased DADs and Ca^{2+} waves in cardiomyocytes carrying the variant.

Therapeutic Insights and Gene Therapy in Animal Models

Transitioning to therapeutic insights, Blackwell et al provided insights into the anatomic origin of CPVT. They focused on genetic manipulation in mice, elucidating that these arrhythmias likely emanate from the Purkinje myocardial junction.⁴⁵ Integrating this discovery into the broader landscape of CPVT research enhances our understanding of the condition's pathogenesis and may open new avenues for targeted therapeutic interventions. Turning to gene therapy, a recent study by Moore et al, the long-term efficacy and safety of cardiac genome editing for CPVT were investigated.⁴⁶ Using an innovative approach with adeno-associated virus type 9-mediated somatic genome editing, the researchers targeted the mutant *RYR2* allele in mice heterozygous for the R176Q variant. Remarkably, this intervention led to a 100% reduction in ventricular tachyarrhythmias compared with controls at 6 weeks after injection, a reduction maintained even after 12 months. In a similar vein, Denegri et al demonstrated that a single delivery of adeno-associated virus type 9-CASQ2 in newborn R33Q mice prevented the development of CPVT and effectively reduced the CPVT phenotype in adult mice.⁴⁷ Innovatively using gene therapy, Santiago Castillo et al studied the potential of cardiac CASQ2 overexpression as a gene therapy treatment for CPVT1. Their results showed a significant reduction in arrhythmias in treated mice compared with untreated ones, proving that CASQ2 overexpression could be an effective and safe therapeutic strategy for CPVT1.⁴⁸ Exploring pharmacological interventions, the study by Ritschel et al investigated the efficacy of the *RYR2*-selective inhibitor ent-(+)-verticilide in preventing ventricular arrhythmias with chronic and oral dosing.⁴⁹

Pharmacological Investigations in Animal Models

Identifying new therapeutic avenues, Reyes Gaido et al identified ruxolitinib as a potent inhibitor of calmodulin-dependent kinase II activity, offering a potential

treatment avenue for cardiac arrhythmias.^{50,51} Through a drug repurposing screen using an improved fluorescent reporter, they effectively found ruxolitinib to inhibit calmodulin-dependent kinase II in cultured cardiomyocytes and mouse models. Assessing drug effectiveness, Hwang et al evaluated the effectiveness of class I antiarrhythmic drugs on *RYR2* channels.⁵² In a CPVT cellular model (*CASQ2*^{-/-} myocytes), both flecainide and propafenone significantly reduced arrhythmogenic calcium waves. In *CASQ2*^{-/-} mice, both propafenone enantiomers effectively prevented exercise-induced bidirectional tachycardia in CPVT cardiomyocytes. The study by Watanabe et al highlighted a specific therapeutic agent and demonstrated the effectiveness of other class 1 antiarrhythmic drugs like flecainide in treating CPVT in both mice and humans.⁵³ It revealed that flecainide inhibits *RYR2* receptor-mediated calcium release. This drug successfully prevented arrhythmias in a CPVT mouse model and completely halted CPVT symptoms in 2 patients who had not responded to conventional treatments. Investigating flecainide's mechanism of action, in the study by Kryshnal et al, the effectiveness of flecainide in treating CPVT was explored.⁵⁴ Using *N*-methylated flecainide analogs, which blocked sodium channels but not *RYR2* channels, they demonstrated that these analogs did not prevent arrhythmias in a CPVT mouse model. The study by Bueno-Levy et al examined ivabradine's effectiveness in treating CPVT by examining ivabradine's efficacy.⁵⁵ The study tested ivabradine on human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) from patients with CPVT, SAN calcium transients, and on ECG measurements from a transgenic mouse model of CPVT. Developing patient-specific models, in the study by Di Pasquale et al., a patient-specific model for CPVT was developed using human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs).⁵⁶ The study demonstrated that treatment with the calmodulin-dependent kinase II inhibitor KN-93 significantly reduced DADs. Additionally, the antiarrhythmic effect of KN-93 was confirmed in these hiPSC-CMs, which paralleled findings in *RYR2* knock-in CPVT mice. For pharmacokinetic analysis, Blackwell et al conducted an in-depth analysis of ent-verticilide's pharmacokinetic and pharmacodynamic properties in a murine model of CPVT. Ent-verticilide is an enantiomer of verticilide, a selective and potent inhibitor of *RYR2*. In their study, Blackwell et al found that ent-verticilide exhibits enhanced stability in plasma compared with its natural counterpart, nat-verticilide, making it more suitable for pharmacological applications.⁵⁷ Adding to these advancements, Liu et al conducted an in vivo study on CPVT using a calsequestrin-associated mouse model. Their research centered on engineering a calmodulin protein to normalize the *RYR2* refractory period, which is crucial

in CPVT development.³⁰ Despite the invaluable insights provided by animal models (Figure 2), their limitations in statistical power and relevance to human physiology should be emphasized. The differences in cardiac physiology between animals and humans can affect the translation of findings.⁵⁸ Also, many animal studies are conducted with relatively small sample sizes, which may compromise the statistical significance of the observed effects. This is particularly important for studies focused on gene therapy interventions, where the variability in genetic backgrounds between animal subjects can further complicate the interpretation.⁵⁹

hiPSC-CM MODEL

Human-Induced Pluripotent Stem Cell-Derived Cardiomyocytes

In vitro models are important for the study of CPVT. These models include mainly induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs) in 2-dimensional models and 3-dimensional cardiac tissue models. iPSCs-CMs are particularly significant as they are generated from patient-specific iPSCs, carrying the same genetic variants, which allows for the exploration of disease mechanisms and drug responses in a personalized context (Figure 2). Genome-edited iPSC lines, often modified using techniques like clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9, are used to induce specific genetic variants and to study the phenotype. In another manner an available variant can be corrected using CRISPR/Cas9 to reverse the phenotype. The recent development of 3-dimensional cardiac tissue models offers a more physiologically relevant environment for studying cardiac channelopathies. The advantage of 3-dimensional cardiac tissue models lies in their ability to better mimic the in vivo microenvironment of the heart, including cell-cell and cell-matrix interactions, mechanical properties, and chemical signaling, compared with traditional 2-dimensional culture models.⁶⁰

hiPSC-CM Model for Precision Medicine

These in vitro models are essential in elucidating the cellular and molecular underpinnings of CPVT and in screening potential pharmacological treatments, enhancing our understanding and management of CPVT. This method has yielded crucial cardiomyocyte models tailored to individual patients, uncovering detailed molecular anomalies and ion channel behavior essential to CPVT. This advancement has driven the creation of customized therapeutic solutions, advancing precision medicine in treating this genetic disorder. Many studies, as listed in Table 2, highlight the significance of hiPSCs in CPVT research, demonstrating their extensive potential as in vitro models. Li et al's research

confirms that CPVT-hiPSCs retain a normal karyotype, thus affirming their accuracy in simulating cardiomyocyte pathology linked to CPVT. Although maintaining a normal karyotype is a fundamental criterion, it ensures that the genetic integrity of the hiPSCs remains preserved, free from chromosomal abnormalities.⁶¹ Jung et al produced CPVT patient-specific iPSC-CMs with the *RYR2* S406L variant. These cells showed elevated diastolic Ca^{2+} levels, reduced SR Ca^{2+} content, and increased arrhythmic risk under stress, indicating faulty Ca^{2+} regulation.⁶² Pölönen et al studied CPVT-hiPSC-CMs with *RYR2* E3D and L4115F mutations, finding unique action potential characteristics under adrenaline that led to pronounced arrhythmic tendencies.⁶³ Carvedilol and flecainide reduced arrhythmic impacts but also decreased cell contraction amplitude. Itzhaki et al developed CPVT-hiPSCs from dermal fibroblasts of a patient with a unique M4109R *RyR2* mutation.⁶⁴ These cells were differentiated into cardiomyocytes and compared with those from healthy controls, revealing significant internal Ca^{2+} handling abnormalities in CPVT cardiomyocytes, contributing to CPVT's pathological manifestations. Similarly, Preininger et al used hiPSC-CMs to study CPVT, focusing on a patient with a unique *RYR2* mutation who responded poorly to nadolol but well to flecainide.⁶⁵ The hiPSC-CMs showed altered Ca^{2+} homeostasis. Nadolol failed to correct this, while flecainide was effective, matching the patient's response. Both studies highlight the value of hiPSC-CMs in modeling CPVT, providing insights into arrhythmic processes and drug evaluations. CPVT arises from dysfunctional intracellular Ca^{2+} regulation due to mutations in either the *RYR2* or *CASQ2* genes. Novak et al and Nijak et al found that mutated *CASQ2* and *RYR2* hiPSC-CMs were less developed and had atypical responses to isoproterenol, causing arrhythmias or increased diastolic $[\text{Ca}^{2+}]_i$, unlike controls.^{66,67} This underscores hiPSC-CMs' role in understanding the distinct effects of *RYR2* and *CASQ2* mutations in CPVT. Maizels et al used hiPSCs from a CPVT2 patient with a D307H-*CASQ2* mutation and observed irregularities in Ca^{2+} handling and an increased susceptibility to arrhythmias.⁶⁸ They tested several drugs, with carvedilol and JTV-519 stabilizing the SR and preventing arrhythmias. In a study, Gao et al found that hiPSC-CMs from a CPVT patient with the *CALM2* p.E46K variant had severe electrical and calcium handling issues due to increased Ca^{2+} leakage through *RYR2*.⁶⁹ The E46K-calmodulin variant strongly enhanced *RYR2* function. Nadolol and flecainide effectively reduced these abnormalities, suggesting potential treatments. The ongoing research utilizing hiPSC-CMs continues to demonstrate considerable promise for developing personalized treatments and further enhancing understanding in the CPVT field. hiPSC-CMs effectively model the CPVT2 phenotype, offering valuable insights into the

Table 2. Summarizes Preclinical Studies on CPVT Syndrome

References	Organism/origin	Gene	Variant	Pharmacological/biological therapeutic interventions
Mohamed et al ³⁷	Mouse	<i>RYR2</i>	-	Testing β blockers to prevent arrhythmias
Veress et al ³⁸	Rat	<i>Caln2</i> <i>Caln3</i>	D132E	β -Adrenergic blockade to manage ventricular tachycardia; potential interventions targeting both <i>RYR2</i> and LTCC dysfunction caused by impaired calmodulin interactions
Deb et al ³⁹	Mouse	<i>CASQ2</i>	Knockout	Modulation of mitochondrial phosphate transport to enhance Ca buffering via inhibitors, cytosolic phosphate adjustment, or phosphate transporter overexpression
Horváth et al ³²	Rabbit (male, female, dihydrotestosterone-treated females)	-	-	Dihydrotestosterone treatment to enhance L-type Ca^{2+} current density; investigation of forward $\text{Na}^+/\text{Ca}^{2+}$ exchanger and calcium handling to assess sex-based differences in arrhythmogenesis
Lin et al ⁴⁰	Mouse	<i>TECRL</i>	Knockout	Targeting fatty acid metabolism and calcium-handling pathways; potential inhibition of calmodulin-dependent kinase II and modulation of metabolic processes
Pavlaki et al ⁴¹	Mouse	-	-	Targeting β -adrenergic receptor signaling via adeno-associated virus type 9-mediated delivery or CRISPR/CAS9-AAV9-based somatic mutagenesis (CASA AV) to disrupt LTCC regulation and calmodulin-dependent kinase II hyperactivity
Neco et al ⁴²	Mouse	<i>RYR2</i>	R4496C	β Blockers, sodium channel blockers (flecainide, propafenone) to manage CPVT; investigation of SAN dysfunction and Ca^{2+} -handling impairments
Yin et al ⁴⁴	hiPSC-CMs and mouse (knock-in) models	<i>RYR2</i>	R420Q	β Blockers and stress management for CPVT
Blackwell et al ⁴⁵	Mouse	<i>CASQ2</i>	-	Endocardial ablation or restoring <i>CASQ2</i> expression in subendocardial cardiomyocytes near Purkinje cells to prevent arrhythmias
Moore et al ⁴⁶	Mouse	<i>RyR2</i>	R176Q	Adeno-associated virus type 9-mediated CRISPR/Cas9 genome editing to specifically disrupt the mutant <i>RYR2</i> allele, preventing arrhythmias while preserving normal cardiac function
Denegri et al ⁴⁷	Mouse	<i>CASQ2</i>	p.Arg33Gln (R33Q)	Adeno-associated virus type 9-mediated gene transfer of the wild-type <i>CASQ2</i> gene, restoring normal calcium handling and interaction with junctin and triadin, effectively preventing and reversing CPVT phenotype
Santiago Castillo et al ⁴⁸	Mouse	<i>RYR2</i>	p.Arg4496Cys (R4496C)	Adeno-associated virus type 9-mediated overexpression of <i>CASQ2</i> , reducing arrhythmic events by promoting <i>RYR2</i> channel closure
Ritschel et al ⁴⁹	Mouse	<i>RYR2</i>	-	Chronic and oral administration of ent-(+)-verticilide
Reyes Gaido et al ^{50,51}	Mouse	<i>CaMKII</i>	-	Therapeutic strategies targeting calmodulin-dependent kinase II
Hwang et al ⁵²	Mouse	<i>RYR2</i> and <i>CASQ2</i>	-	Flecainide and propafenone shown to prevent CPVT
Watanabe et al ⁵³	Mouse	<i>RYR2</i>	-	Flecainide demonstrated to inhibit <i>RYR2</i> -mediated Ca^{2+} release and prevent CPVT in mouse models
Kryshtal et al ⁵⁴	Mouse	<i>RYR2</i>	-	Flecainide inhibits <i>RYR2</i> -mediated Ca release, the primary antiarrhythmic mechanism in CPVT
Bueno-Levy et al ⁵⁵	Mouse	<i>CASQ2</i>	D307H	Ivabradine was ineffective in preventing arrhythmias
Blackwell et al ⁵⁷	Mouse	<i>RyR2</i>	-	ent-verticilide showed dose-dependent antiarrhythmic efficacy with favorable pharmacokinetics
Devalia et al ²¹	CPVT-patient iPSC-CMs	<i>TECRL</i>	p. Arg196Gln c.331 + 1G > A	Flecainide significantly reduced triggered activity in <i>TECRL</i> -mutant hiPSC-CMs
Jung et al ⁶²	CPVT-patient iPSC-CMs	<i>RYR2</i>	p.(Ser406Leu) p.(Pro2328Ser)	Dantrolene restored normal Ca^{2+} handling and prevented arrhythmias
Itzhaki et al ⁶⁴	CPVT-patient iPSC-CMs	<i>RYR2</i>	p.(Met4109Arg)	Adrenergic blockers, flecainide, and thapsigargin reduced afterdepolarizations
Preininger et al ⁶⁵	CPVT-patient iPSC-CMs	<i>RYR2</i>	p. (Leu3741Pro)	β Blockers (nadolol), flecainide; assessed response to adrenergic stimulation
Novak et al ⁶⁷	CPVT-patient iPSC-CMs	<i>RYR2</i>	p. (Ile4587Val)	Investigated effects of isoproterenol, caffeine, and ryanodine on calcium handling

(Continued)

Table 2. Continued

References	Organism/origin	Gene	Variant	Pharmacological/biological therapeutic interventions
Novak et al ⁶⁷ and Nijak et al ⁶⁶	CPVT-patient iPSC-CMs	CASQ2	p. (Asp307His)	Explored ultrastructural and functional differences between CPVT1 and CPVT2 mutations
Pölonen et al ⁶³	CPVT-patient iPSC-CMs	<i>RYR2</i>	Exon 3 deletion (E3D), L4115F	Tested carvedilol and flecainide; carvedilol resolved L4115F arrhythmias
Maizels et al ⁶⁸	CPVT-patient iPSC-CMs	CASQ2	p. (Asp307His)	Antiarrhythmic agents including β blockers, flecainide, riluzole showed efficacy, carvedilol stabilizing sarcoplasmic reticulum function
Gao et al ⁶⁹	CPVT-patient iPSC-CMs	<i>CALM2</i>	p.E46K	Nadolol and flecainide were effective in suppressing abnormal Ca^{2+} waves caused by the E46K variant in RyR2-associated arrhythmogenicity
Li et al ⁶¹	CPVT-patient iPSC-CMs	<i>RYR2</i>	p.A2254V	Generated hiPSC lines for disease modeling and studying CPVT mechanisms

CASQ2 indicates cardiac calsequestrin 2; CPVT, catecholaminergic polymorphic ventricular tachycardia; CRISPR, clustered regularly interspaced short palindromic repeats; hiPSC-CM, human-induced pluripotent stem cell–derived cardiomyocyte; iPSC-CM, induced pluripotent stem cell–derived cardiomyocyte; LTCC, L-type calcium channel; SAN, sinoatrial node; and TECRL, trans-2,3-enoyl-CoA reductase–like protein.

disease's mechanisms and potential patient-specific treatments.⁶⁸ Overall, the ability of hiPSC-CMs to accurately model patient-specific cardiomyocyte pathology has been crucial in understanding CPVT and exploring potential therapeutic interventions.⁶⁸

Progress in Tissue Engineering Using hiPSC-CMs

The advent of 3-dimensional engineered heart tissues (EHTs) utilizing hiPSC-CMs has significantly advanced the research on conditions like CPVT. These developments have effectively tackled the issue of hiPSC-CMs' immaturity, leading to substantial enhancements in their functionality and structure. Approaches to promote cardiomyocyte maturity have been crucial. These strategies involve extended culture durations, the addition of thyroid hormone, and incorporating fatty acids as energy sources. Additionally, coculture methods, using extracellular matrix, and applying mechanical or electrical stimulation in 3-dimensional cultures have been instrumental in achieving cardiomyocyte maturation.⁷⁰ The heart comprises various celltypes, not just cardiomyocytes. Studies show that cardiomyocytes represent only about 25% to 35% of the heart's cells, while endothelial cells account for 60%, and fibroblasts are <20%.⁷¹ In specific heart regions, such as the atria and ventricles, the distribution changes, with cardiomyocytes making up 30% to 50% of cells, and endothelial cells and fibroblasts contributing 10% and 20%, respectively.⁷² Incorporating this diversity of cell types enhances the physiological relevance of these models, making them more applicable in disease study, drug testing, and cardiotoxicity assessments. The scaffold-free hanging droplet technique for 3-dimensional cell culture involves placing iPSC-CMs in droplets on ultra-low attachment plates, sometimes adding cardiac fibroblasts and endothelial cells.⁷³ This technique produces stable 3-dimensional iPSC-CM

models responsive to various stimuli. However, challenges like limited Ca^{2+} dye penetration and retention of immature characteristics, such as shorter sarcomeres, are observed.⁷³ In addition, the absence of hormonal and nonhormonal factors is a very important limitation of the 3-dimensional model. Combining iPSC-CMs with human cardiac fibroblasts and coronary artery endothelial cells results in cardiac spheroids that more accurately resemble the in vivo human heart.⁷⁴ Despite these advancements, substantial differences limit the utility of even the best 3-dimensional models, particularly in replicating the full complexity of native heart tissue. Cardiac microtissues, another type of scaffold-free product, are created by mixing different cell types, including iPSC-derived endothelial cells, cardiac fibroblasts, and iPSC-CMs. These cardiac microtissues show mature structures like elongated tubular myofibrils and T-tubule–like structures, with electrophysiological maturity indicated by action potential notches, effectively replicating disease phenotypes.⁷⁵ In scaffold-based 3-dimensional cultures, scaffolds made from decellularized extracellular matrix or natural or synthetic polymers are used.⁷⁶ These scaffolds aid in the maturation of cardiomyocytes and using adult heart extracellular matrix has been shown to enhance maturation in both 2D and 3D cultures.⁷⁷ EHTs, where iPSC-CMs are cultured on hydrogel scaffolds, display more mature electrophysiological properties and cardiac ultrastructure, aligning more closely with adult cardiac tissue.⁷⁸ Progressive stretching of EHTs has been used to further advance maturation, leading to improved contractility and a more mature excitation/contraction coupling. These EHT models are invaluable in channelopathy research, showing less arrhythmogenic activity compared with 2-dimensional models and better mimicking clinical scenarios.⁷⁹ Specifically, in CPVT research, 3-dimensional models like EHTs are proving to be essential in understanding the pathophysiological mechanisms of arrhythmias

by allowing for more accurate modeling of calcium-handling abnormalities and the effects of β -adrenergic stimulation. These models also facilitate the development and testing of potential therapeutic interventions in a more in vivo–like environment compared with traditional 2-dimensional cell cultures.⁸⁰

Limitations of hiPSC-CMs

While the development of iPSC technology has significantly advanced disease modeling, it still faces notable limitations. A primary issue is the immaturity and variability of hiPSC-CMs, which manifest in altered gene expression, poor sarcomere organization, abnormal morphology, and insufficient T-tubule formation.⁸¹ These immature characteristics result in less stability and an increased risk of arrhythmic events in hiPSC-CMs compared with adult cardiomyocytes, reflecting an inadequate replication of disease-specific phenotypes found in CPVT.⁸² The hiPSC-CMs often do not account for critical factors such as cellular interactions and hormonal regulation in single-cell studies.⁸³ Efforts to address their immature phenotype include extending culture times, using nutrient-rich media, and using various stimulations such as electrical fields and mechanical forces within extracellular matrices and 3-dimensional cardiac tissues.⁸⁴ Moreover, hiPSC-CMs lack the complexity of natural cardiac tissues, missing essential ion channels and cytoskeleton proteins. Similarly to animal models, hiPSC-CMs also face challenges regarding their statistical robustness. The variability in hiPSC-CMs differentiation processes and the immaturity of these cells may introduce inconsistencies in experimental results. Moreover, small sample sizes in hiPSCs studies can lead to issues with statistical power and impact the ability to draw a final conclusion about the pathophysiology and treatment responses.⁸⁵ Advances such as engineered heart tissues and organ-on-a-chip technologies aim to enhance the structural complexity and improve the predictive power of these models.⁸⁶ While hiPSC-CMs have proven valuable for drug development and cardiovascular research, their potential as a full human model is restricted due to their disparities with adult human cardiomyocytes.⁸³ The complexity of genetic interactions and environmental factors in disease manifestation remains difficult to model. Even with advanced genome editing tools like CRISPR/Cas9, challenges persist.⁸⁷ Overall, continued research and improvements are needed to address these limitations, with ongoing developments showing promise in enhancing the capabilities and applications of hiPSC technology.

Unanswered Questions and Future Research

Like the “dark side of the moon,” some areas of CPVT remain challenging to study directly, and clinicians

must often make decisions with limited evidence. One major area for further research is understanding the full spectrum, including pathogenicity of genetic mutations beyond the commonly studied *RYR2* and *CASQ2* variants. These mutations often overlap with other cardiomyopathies and channelopathies, suggesting that CPVT may not be confined to the heart alone but could involve multiorgan systems.⁸⁸ Additionally, recent findings emphasize the important role of the autonomic nervous system in cardiac dysregulation, where structures like the stellate ganglion contribute to arrhythmogenic mechanisms via neuroimmune pathways. Studies on stellate ganglion function reveal its significant influence on cardiac sympathetic nerve activity, particularly in arrhythmias involving inflammation. Abnormal stellate ganglion activity may provoke sympathetic overexcitation, initiating a cycle of inflammation and arrhythmias. Exploring interventions targeting stellate ganglion function and inflammatory responses may present new strategies for preventing CPVT-related arrhythmias and enhancing treatment options for cardiovascular dysautonomia.⁸⁹ Many studies have shown that CPVT overlaps with neurodevelopmental disorders,⁹⁰ particularly intellectual disabilities, indicating that shared pathophysiological mechanisms may involve intracellular calcium handling in both the heart and the brain.⁹⁰ Recently, Tonko et al emphasized that autonomic triggers play a critical role in the proarrhythmogenic mechanisms of cardiac channelopathies, including CPVT.⁹¹ Their findings suggest that autonomic regulation, potentially involving Purkinje fiber activity, could be modulated using neuromodulation approaches. By targeting specific G-protein–coupled receptors, neuropeptides, and central nervous system centers responsible for autonomic control, these emerging strategies may help reduce Purkinje-mediated ventricular arrhythmias in CPVT. Future studies should explore the role of Purkinje fibers in human CPVT and assess whether ablation techniques may be effective in certain cases. Additionally, the overlap between CPVT and other cardiomyopathies, such as arrhythmogenic right ventricular cardiomyopathy, is not fully understood and requires further investigation to clarify diagnosis and treatment.⁹² The role of the implantable cardioverter-defibrillator in CPVT remains controversial, as recent evidence suggests it may not provide a mortality benefit in all cases. This raises important questions about the appropriate criteria for implantable cardioverter-defibrillator implantation, especially in patients who have not experienced life-threatening arrhythmias.⁹³ Treatment modalities also need refinement. While nonselective β blockers like nadolol and propranolol remain the cornerstone of therapy, many patients require additional interventions such as flecainide or left cardiac sympathetic denervation.⁹⁴ In addition, whether several CPVT types may

have a variable response to drug treatment has not been systematically studied yet. Pharmacological strategies have expanded in recent years, but the relative efficacy of different treatments, particularly for asymptomatic carriers identified through genetic screening, remains underexplored. Exploring new therapeutic options, including advanced gene-editing approaches, will be crucial in the coming years, particularly for patients with complex CPVT phenotypes. Given the increasing evidence linking certain RYR2 to seizures,⁹⁵ it will also be important to consider a multidisciplinary approach to treatment, integrating insights from neurology. Unanswered questions in CPVT research also include the role of common and rare variants both in disease genes and other genes, epigenetic variation, and other changes that influence the pathogenicity of presumed disease-causing mutations. Ultimately, collaborative efforts between geneticists, clinicians, and bioengineers will be crucial for addressing these unanswered questions, refining our understanding of CPVT, and developing personalized treatment approaches.

CONCLUSIONS

The exploration of CPVT through diverse experimental models has significantly enriched our understanding of its pathophysiology and potential treatment strategies. Animal models have provided essential insights into the systemic physiological and molecular dynamics of CPVT, while hiPSC-CMs have facilitated detailed genotype–phenotype analyses and the evaluation of personalized therapeutic interventions. These models, especially when integrated with advanced technologies such as CRISPR/Cas9 and 3-dimensional cardiac tissues, have substantially advanced our capability to replicate human cardiac pathology and assess pharmacological responses effectively. Despite their transformative impacts, each model carries inherent limitations that highlight the need for continuous refinement. The collective insights from these models drive the development of targeted therapies, aiming to reduce the clinical burden of this potentially fatal condition.

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